

N+- Network security, Maintenance and Troubleshooting procedures

Build a SOHO network

[Q-1] Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

[Broadband ISP modem] --wired (Ethernet)--> [Wi-Fi Router (AX55)]

| --wired--> [Printer] (static IP)

| --wired--> [5-port Gigabit Switch] --> (optional wired devices)

| --wireless--> [Laptop A]

| --wireless--> [Laptop B]

| --wireless--> [Smart TV] (static IP)

Connections labelled: modem router (wired); router printer (wired) or (wireless); laptops & TV (wireless); optional switch (wired).

[Q-2] List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Hardware components

- **Wi-Fi router:** *TP-Link Archer AX55 (AX3000)* — Wi-Fi 6, good value for SOHO.
- **Smart TV (43"):** *Xiaomi X Pro 43" or Samsung Vision AI 43"* (choose based on budget).
- **Printer:** *Epson Eco Tank L3250 or Canon PIXMA Mega Tank G3010* (Wi-Fi ink-tank models).
- **Unmanaged switch (if >1 wired device):** *TP-Link TL-SG105 5-port Gigabit switch*.
- **Ethernet cables:** **Cat6** patch cables (1–10 m as needed).
- **Power strip, wall mounts, cable ties.** (*All models widely available on Amazon/Flipkart.*)

[Q-3] Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Router configuration steps (concise)

1. **Connect** modem WAN → router WAN; power on.
2. Open browser to **192.168.0.1** (or router label) and log in with default credentials.
3. **Change admin password** immediately to a strong passphrase. **Save**.
4. Under Wireless settings: set **SSID** to a unique name (e.g., SuratSOHO_5G and SuratSOHO_2G).
5. Set **Security** to **WPA2-PSK (AES)** (or WPA3 if all clients support it); choose a strong Wi-Fi password.
6. Enable **DHCP** on router for client IP assignment; set DHCP pool (e.g., 192.168.1.100–192.168.1.200).
7. Reserve static IPs (DHCP reservation) for **printer** and **TV** or configure them on the devices.
8. **Save** and reboot router; test connectivity from a laptop.

[Q-4] Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	MAC address	IP address
Router (LAN)	00:11:22: AA: AA:01	192.168.1.1
Printer (static)	00:11:22: AA:BB:10	192.168.1.10
Smart TV (static)	00:11:22: AA:BB:20	192.168.1.20
Laptop A (DHCP)	00:11:22: AA:CC:30	DHCP (e.g., 192.168.1.101)
Laptop B (DHCP)	00:11:22: AA:CC:31	DHCP (e.g., 192.168.1.102)
Switch	N/A (unmanaged)	N/A

[Q-5] Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Possible cause 1 — Wi-Fi authentication issue.

- *Diagnose:* Can the laptop see SSID? Try reconnecting; check Wi-Fi password.
- *Fix:* Forget network → rejoin with correct password; check router's MAC filter or client limit.

Possible cause 2 — DHCP or IP conflict.

- *Diagnose:* Run ipconfig/ifconfig to see IP; if APIPA (169.254.x.x) or conflict, DHCP failed.
- *Fix:* Reboot laptop; release/renew DHCP; check router DHCP pool and reservation table.

Possible cause 3 — Router or WAN outage.

- *Diagnose:* Other devices online? Ping router (192.168.1.1). If router reachable but no internet, check modem.
- *Fix:* Reboot modem and router; check ISP status; connect laptop via Ethernet to isolate Wi-Fi vs ISP issue.